Answer Key

1. B
2. C
3. B
4. C
5. B
6. A

Q7

- what happens mathematically if activations are removed,
- why stacking linear layers alone is insufficient,
- and how nonlinear activations improve representation learning.

Q8

- Reconstruction-based methods work by reducing the data to a lower-dimensional representation and then projecting it back into its original space.
- The underlying assumption is that normal instances can be better reconstructed from this compressed space than anomalies.
- Outliers are detected by measuring the "reconstruction error," which is the difference between the original data and the reconstructed data.

Q9

- Pretext tasks may overfit task-specific signals
- Contrastive → general similarity structure

Q10

- Sequential dependency (time step t depends on t−1)
- Cannot parallelize

Q11

- Model can “peek” future tokens
- Leads to unrealistic training
- Poor generalization at inference

Q12

(a) Scores:

- $q \cdot k_1 = 1$
- $q \cdot k_2 = 0$

(b) Softmax:

- exp(1)=e, exp(0)=1
- weights = [e/(e+1), 1/(e+1)]

(c) Output:

$= w_1 \cdot v_1 + w_2 \cdot v_2$

$= [2e/(e+1), 3/(e+1)]$

Q13

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Q14

- RNN: sequential flow

- Transformer: global attention
- Complexity:
  - RNN: O(n)
  - Transformer: $O(n^2)$

Q15

- SSL → pretraining
- Foundation model → large pretrained model
- Fine-tuning → adapt to task

Q16

- shortcut learning
- domain mismatch
- weak pretext tasks

Improvement:

- contrastive learning
- multimodal learning

Q17

- augmentations (cropping, masking, noise)
- contrastive loss
- evaluation via downstream task